

MANAGEMENT REVIEW

Management without theatre.

AI

THE QUALITY OF DATA

How many new records carry an error, what bad data costs, six dimensions of quality, data cascades and wrong labels in AI, and a card for measuring your data in one afternoon.



47%

of newly created records had at least one critical error, on average, in 75 checks that managers ran on their own units.

NAGLE, REDMAN & SAMMON, 2017

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Every report, KPI and AI model is only as good as the records behind it. This issue asks how many of those records are wrong, what that costs, and how a team can find out in one afternoon.

When 75 managers checked the newest records of their own units, 47% had at least one critical error on average, and fewer than 3% of the scores met the bar they set themselves. Estimates of the cost run from Gartner's \$12.9 million a year per organisation to IBM's \$3.1 trillion for the US, neither with a visible method. In AI, 92% of 53 practitioners interviewed had met a data cascade, and the test sets of widely used benchmarks hold at least 3.3% wrong labels on average.

Stiven Janaqi EDITOR · STIVEN CATALYST

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HOW TO READ THIS ISSUE

Figure

Every figure has its source and year at the foot of its page.

Our reading

Where the editors interpret rather than the research, it says so.

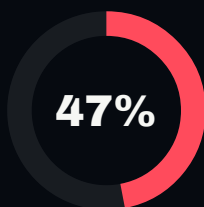
Practice

The steps and the card are proposals to try, not research results.

COVER STORY

47 out of 100 new records

Over two years, 75 managers ran the same test on their own units: they took the last 100 records their teams had created or used and counted those without a critical error, one that affects the work.



of newly created records had at least one critical error, on average.

Asked how good the data had to be, no manager called a score below the high nineties acceptable, and fewer than 3% of the scores reached that bar. The scores ranged from 0 to 99, and the authors found no significant differences between industries.

OUR READING

The test takes an afternoon and needs no software. The hard part is looking at the records your own team made.

Source: Tadgh Nagle, Thomas C. Redman & David Sammon, Harvard Business Review, 2017

75 measurements by managers in their own units: a small sample, not representative. The bar was the managers' own, and the 3% are measurements, not companies.

THE NUMBERS

What bad data costs

Nobody has measured the full cost of poor data. What exists are estimates and rules of thumb, worth reading together with their caveats.

\$12.9m

a year: Gartner's average cost of poor data quality per organisation (2021)

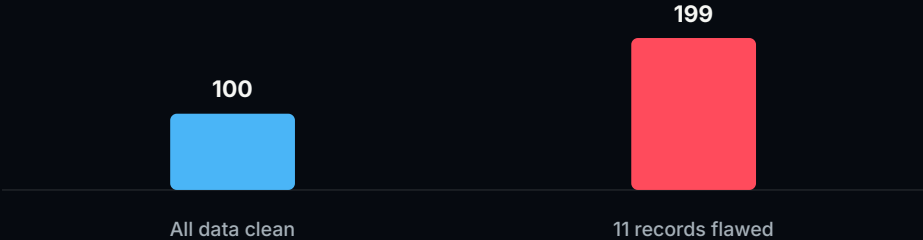
50%

of knowledge workers' time lost in hidden data factories, by Redman's estimate

44%

of German firms that considered AI but do not use it cite the availability or quality of data (2025)

COST OF 100 UNITS OF WORK AT \$1 EACH, BY REDMAN'S RULE OF TEN (\$)



Redman calls the extra steps a team adds to cope with errors made upstream hidden data factories, and his rule of ten says flawed data make a unit of work cost ten times as much. Both are his estimates, not measurements. The best-known figure, \$3.1 trillion a year for the US in 2016, was IBM's; Redman later wrote that it could be off by a trillion dollars either way.

Sources: Manasi Sakpal, Gartner, 2021; Thomas C. Redman, Harvard Business Review, 2016 (via SAP Community, 2023); Statistisches Bundesamt (Destatis), 2025; Tadhg Nagle, Thomas C. Redman & David Sammon, Harvard Business Review, 2017

Estimates, not measurements, and not to be added up or compared. Gartner does not show how it reached its average; the German figure is a reason firms report, not a measure of data quality.

THE MODEL

Six dimensions of quality

In 2013 a working group of DAMA UK, the data management association, named six primary dimensions for assessing data quality. The UK government adopted them in its 2020 data quality framework.

01

Completeness

nothing that should be there is missing

02

Uniqueness

no record appears twice

03

Timeliness

changes at the source show up quickly where the data are used

04

Validity

values fit the agreed format and rules

05

Accuracy

the data reflect reality

06

Consistency

the data agree across sources, which does not make them accurate

For AI, Gartner sets a different test: data are AI-ready when they represent the use case, including the patterns, errors and outliers a model needs. In a 2024 Gartner survey of 1,203 data management leaders, 63% said their organisation did not have, or was not sure it had, the right data practices for AI.

OUR READING

Clean is not the same as fit for use. The question is always: good enough for what?

Sources: DAMA UK Working Group, 2013 (via UK Government Data Quality Framework, 2020); Gartner, 2025

The definitions follow summaries of the DAMA UK document; DAMA-DMBOK and other sources give longer lists. The 63% is a self-report.

[More in the essay](#)

Mistakes get lost between departments

WHAT THE RESEARCH SAYS

Everyone wants the model work

Nithya Sambasivan and colleagues at Google interviewed 53 practitioners who build AI for high-stakes uses in India, East and West Africa and the US. 92% had met at least one data cascade: a data problem that compounds and harms results further down the line.

WRONG LABELS IN THE TEST SETS OF WIDELY USED BENCHMARKS (%)

MNIST



Average of 10 datasets



ImageNet validation set



QuickDraw



Curtis Northcutt, Anish Athalye and Jonas Mueller found at least 3.3% wrong labels on average in the test sets of ten widely used datasets of images, text and audio. Such errors can reverse the ranking of models: on ImageNet, the smaller ResNet-18 beats ResNet-50 if the share of originally mislabelled test examples rises by just 6%.

OUR READING

A model is judged against data. If the data are wrong, so is the judgment.

Sources: Nithya Sambasivan et al., ACM CHI, 2021; Curtis G. Northcutt, Anish Athalye & Jonas Mueller, NeurIPS Datasets and Benchmarks, 2021

Interviews with 53 people, not a representative survey; public test sets, not company data. Versions of the paper give 3.3% or 3.4%.

HOW IT IS MEASURED

The Friday afternoon measurement

The method of Nagle, Redman and Sammon needs no software: a team, an afternoon and the last 100 records it created or used.

1

Take the last 100 records

the ones your unit created or used most recently

2

Pick the 10–15 most important fields

the ones the next step depends on

3

Mark every visible error

record by record, with people who know the data

4

Count the error-free records

that number, from 0 to 100, is the score

HYPOTHETICAL EXAMPLE, 100 CUSTOMER ORDERS

Errors wrong address in 9, no phone number in 14, wrong quantity in 5; 4 orders have two errors

Score 24 orders have an error, 76 are clean: score 76

Rule of ten \$100 of work costs $76 + 24 \times 10 = \$316$

The orders and errors are invented.

Source: Tadhg Nagle, Thomas C. Redman & David Sammon, Harvard Business Review, 2017

The steps follow Nagle, Redman & Sammon (2017); the rule of ten is Redman's rule of thumb; the example is the editors'.

TOOL OF THE ISSUE

The data quality card

One card per check, once a month. Count the errors by field, then sort them: a few fields usually carry most of them.

01 Records checked which 100, from which step, on which date

02 Critical fields the 10–15 fields the next step depends on

03 Errors by field missing, wrong, duplicate or out of date

04 Score error-free records out of 100

05 Where errors start the step or system where most are created

06 Fix at the source one change, who, by when; measure again next month

Sources: Tadhg Nagle, Thomas C. Redman & David Sammon, Harvard Business Review, 2017; DAMA UK Working Group, 2013 (via UK Government Data Quality Framework, 2020)

A practice proposed by the editors, after the Friday afternoon measurement (2017) and the DAMA UK dimensions (2013).

Open the tool: [Pareto 80/20](#)

stivencatalyst.com/tools/pareto/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Only 3% of Companies' Data Meets Basic Quality Standards

Tadhg Nagle, Thomas C. Redman & David Sammon, Harvard Business Review, 2017

<https://hbr.org/2017/09/only-3-of-companies-data-meets-basic-quality-standards>

Gründe gegen die Nutzung von Technologien der künstlichen Intelligenz nach Beschäftigtengrößeklassen

Statistisches Bundesamt (Destatis), 2025

<https://www.destatis.de/DE/Themen/Branchen-Unternehmen/Unternehmen/IKT-in-Unternehmen-IKT-Branche/Tabellen/ikti-gegen-nutzung-kuenstliche-intelligenz.html>

"Everyone wants to do the model work, not the data work": Data Cascades in High-Stakes AI

Nithya Sambasivan et al., ACM CHI, 2021

<https://research.google/pubs/everyone-wants-to-do-the-model-work-not-the-data-work-data-cascades-in-high-stakes-ai/>

Pervasive Label Errors in Test Sets Destabilize Machine Learning Benchmarks

Curtis G. Northcutt, Anish Athalye & Jonas Mueller, NeurIPS Datasets and Benchmarks, 2021

<https://arxiv.org/abs/2103.14749>

Bad Data Costs the U.S. \$3 Trillion Per Year

Thomas C. Redman, Harvard Business Review, 2016 · via SAP Community, 2023

<https://hbr.org/2016/09/bad-data-costs-the-u-s-3-trillion-per-year>

Lack of AI-Ready Data Puts AI Projects at Risk

Gartner, 2025

<https://www.gartner.com/en/newsroom/press-releases/2025-02-26-lack-of-ai-ready-data-puts-ai-projects-at-risk>

How to Improve Your Data Quality

Manasi Sakpal, Gartner, 2021

<https://www.gartner.com/smarterwithgartner/how-to-improve-your-data-quality>

The Six Primary Dimensions for Data Quality Assessment

DAMA UK Working Group, 2013 · via UK Government Data Quality Framework, 2020

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked "via". The editors' interpretation is marked "Our reading". Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

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Every issue, one management question, checked against the best research.

ALL ISSUES

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